
Uni DB1 Syntax Details

Syntax details for the DB1 (databases) course at HdM
Stuttgart

Felicitas Pojtinger

2022-02-01

Inhaltsverzeichnis

1	Acknowledgements	3
2	Reset Everything	3
3	SQL	3
3.1	Operators	3
3.2	Joins	4
3.3	Aliases	6
3.4	Limits and Pagination	6
3.5	Dates and Intervals	7
3.6	Expressions	8
3.7	Grouping and Ordering	9
3.8	Counting and Sums	10
3.9	Inserting	10
3.10	Switches	11
3.11	Helper Functions	11
3.12	Auto-Generated Primary Keys	12
3.13	Modifying Columns	12
3.14	Virtual Columns	13
3.15	Modifying Tables	13
3.16	Constraints	14
3.17	Types	15
3.18	Views	15
3.19	Indexes	16
4	PL/SQL	16
4.1	Block Structure	16
4.2	Variables	17
4.3	Fetching Data	18
4.4	Branches and Expressions	18
4.5	Switches	19
4.6	Labels and Goto	19
4.7	Loops	20
4.8	Types and Objects	21
4.9	Exceptions	21
4.10	Cursors	22
4.11	Locks	23

4.12 Procedures	23
4.13 Functions	24
4.14 Packages	25
4.15 Triggers	26
4.16 Maps	29
4.17 Arrays	29

“so basically i am monkey” - monke, *monkeeee*

1 Acknowledgements

Most of the following is based on the [Oracle Tutorial](#).

2 Reset Everything

Run the following to get the commands to drop all tables and their constraints:

```
1 begin
2   for i in (select index_name from user_indexes where index_name not
3             like '%_PK') loop
4     execute immediate 'drop index ' || i.index_name;
5   end loop;
6   for i in (select trigger_name from user_triggers) loop
7     execute immediate 'drop trigger ' || i.trigger_name;
8   end loop;
9   for i in (select view_name from user_views) loop
10    execute immediate 'drop view ' || i.view_name;
11  end loop;
12  for i in (select table_name from user_tables) loop
13    execute immediate 'drop table ' || i.table_name || ' cascade
14                      constraints';
15  end loop;
16  execute immediate 'purge recyclebin';
17 end;
```

Now copy & paste the output into SQL Developer's SQL worksheet and hit F5.

3 SQL

3.1 Operators

Operator	Description
=	Equality
!=,<>	Inequality
>	Greater than
<	Less than
>=	Greater than or equal to
<=	Less than or equal to
IN	Equal to any value in a list of values
ANY/ SOME/ ALL	Compare a value to a list or subquery. It must be preceded by another operator such as =, >, <.
NOT IN	Not equal to any value in a list of values
[NOT] BETWEEN n and m	Equivalent to [Not] >= n and <= y.
[NOT] EXISTS	Return true if subquery returns at least one row
IS [NOT] NULL	NULL test

3.2 Joins

- An **inner join** matches stuff in both tables:

```
1 select a.id as id_a, a.color as color_a, b.id as id_b, b.color as  
   color_b from palette_a a inner join palette_b b on a.color = b.  
   color;
```

- A **left (outer) join** matches everything in the left tables plus what matches in the right table:

```
1 select a.id as id_a, a.color as color_a, b.id as id_b, b.color as  
   color_b from palette_a a left join palette_b b on a.color = b.  
   color
```

- This **left (outer) join** matches everything that is in the left table and not in the right table:

```
1 select a.id as id_a, a.color as color_a, b.id as id_b, b.color as  
   color_b from palette_a a left join palette_b b on a.color = b.  
   color where b.id is null
```

- A **right (outer) join** matches everything in the right join plus what matches in the left table:

```
1 select a.id as id_a, a.color as color_a, b.id as id_b, b.color as color_b from palette_a a right join palette_b b on a.color = b.color;
```

- This **right (outer) join** matches everything that is in the right table and not in the left table:

```
1 select a.id as id_a, a.color as color_a, b.id as id_b, b.color as color_b from palette_a a right join palette_b b on a.color = b.color where a.id is null;
```

- A **full (outer) join** merges both tables:

```
1 select a.id as id_a, a.color as color_a, b.id as id_b, b.color as color_b from palette_a a full join palette_b b on a.color = b.color;
```

- This **full (outer) join** merges both tables and removes those rows which are in both:

```
1 select a.id as id_a, a.color as color_a, b.id as id_b, b.color as color_b from palette_a a full join palette_b b on a.color = b.color where a.id is null or b.id is null;
```

- In addition to the on keyword you can also use the using keyword if the PK and FK are the same:

```
1 select * from orders inner join order_items using(order_id)
```

- You can also use multiple on or using statements:

```
1 select * from orders inner join order_items using(order_id) inner join customers using(customer_id)
```

- If you use the on keyword, use **and** for multiples!
- You can also create the Cartesian product:

```
1 select * from products cross join warehouse;
```

- It is also possible to do a self join:

```
1 select (w.first_name || ' ' || w.last_name) "Worker", (m.first_name || ' ' || m.last_name) "Manager", w.job_title from employees w left join employees m on w.employee_id = m.manager_id
```

- What is the difference between **join** and **union**? **join** merges horizontally (there are more columns than before, maybe also more rows), **union** merges vertically (there are more rows than before, but the column count stays the same).

- `union` is similar to `T1 | T2` in TypeScript; you can use `order by` and `union` to remove duplicates, but note that we have to use `select` two times:

```
1 select first_name, last_name, email, 'contact' as role from
   contacts union select first_name, last_name, email, 'employee'
   as role from employees order by role
```

- `union all` is similar to `T1 & T2` in TypeScript; it keeps duplicates:

```
1 select last_name from contacts union all select last_name from
   employees;
```

- Wish to find the difference between two tables? Use `intersect`:

```
1 select last_name from contacts intersect select last_name from
   employees;
```

- Wish to subtract one table from another table? Use `minus`:

```
1 select last_name from contacts minus select last_name from
   employees;
```

3.3 Aliases

- You can alias long column names with `select mylongname as name from contacts` or just `select mylongname name from contacts`. The `as` keyword is optional. Full-text column names are supported by enclosing in `""`. `as` can also format strings: `select first_name || ' ' || last_name as "Name" from employees`; yields Alice, Bob and System.
- You can also create a table alias (using `from employees e`), but you CAN'T USE the `as` keyword.

3.4 Limits and Pagination

- The Oracle equivalent of filter is `fetch n next rows only`: `select * from products order by list_price desc fetch next 5 rows only`;
- You may also use the `fetch next n percent rows only`:

```
1 select * from inventories order by quantity desc fetch next 10
   percent rows only;
```

- Filtering by for example a quantity, and you only want the first 10 “condition matches”? Use `fetch n next rows with ties`:

```
1 select * from inventories order by quantity desc fetch next 5 rows
  with ties;
```

- Need Pagination? Use offset:

```
1 select * from products order by standard_cost desc offset 10 rows
  fetch next 10 rows only;.
```

3.5 Dates and Intervals

- Want to extract a year from a date? Use `extract`:

```
1 select * from orders where status = 'Shipped' and extract(year
  from order_date) = 2017 order by order_date desc fetch next 1
  rows with ties;
```

- Want to get the current date? Use `current_date`:

```
1 select current_date from dual;
```

- The `to_char` function can convert dates (and timestamps) to chars:

```
1 select to_char(sysdate, 'YYYY-MM-DD') from dual;
```

- The `to_date` function can convert chars to dates:

```
1 select to_date('2021-01-12', 'YYYY-MM-DD') from dual;
```

- Alternatively, the date literal uses the `YYYY-MM-DD` format and does not require format specs:

```
1 select date '1969-04-20' from dual;
```

- You can get the current date with `sysdate`:

```
1 select localtimestamp from dual;
```

- You can get the current date & time with `datelocaltimestamp`:

```
1 select localtimestamp from dual;
```

- The current time zone is available with `sessiontimezone`:

```
1 select sessiontimezone from dual (yields Europe/Berlin);
```

- The `timestamp` literal uses the `YYYY-MM-DD HH24:MI:SS.FF` format:

```
1 select timestamp '1969-04-20 00:00:00.00' from dual;
```

- You may also append the timezone (But keep in mind that timestamp with time zone is the column type in this case):

```
1 select timestamp '1969-04-20 00:00:00.00 Europe/Berlin' from dual;
```

- The `interval` literal can be used to create intervals:

```
1 select interval '9' day from dual, select interval '9' month from
dual, select interval '9-2' year to month from dual or select
interval '09:08:6.75' hour to second(2) from dual;
```

- Using the `months_between` function, the count of months between two dates can be computed.

3.6 Expressions

- Only single quotes are supported.
- Comparisons are done with `=`, NOT `==`.
- It also supports full expression evaluation:

```
1 select product_name as "Product Name", list_price - standard_cost as "
Gross Profit" from products order by "Gross Profit"
```

- You can use `()` in `where` clauses to prioritize:

```
1 select * from orders where (
2 status = 'Canceled' or status = 'Pending' ) and customer_id = 44
3 order by order_date;
```

- The `in` keyword is a useful tool for sub collections and subqueries:
 - `select * from orders where salesman_id in (54, 55, 56) order by order_id;`
 - `select * from orders where salesman_id not in (54, 55, 56) order by order_id;` (you can use `not`)
 - `select * from employees where employee_id in ( select distinct salesman_id from orders where status = 'Canceled' ) order by first_name;` (you can of course also use `not`)
- `between` can also be used for dates:

```
1 select * from orders where order_date between date '2016-12-01'
and date '2016-12-31'
```

- `... like '%Asus%'` (note the 's') is basically a full-text search.

- Some examples of `like` (you can use `not` for all of them):
 - `select * from contacts where last_name like 'St%'`
 - `select * from contacts where last_name like '%St'`
 - `select * from contacts where last_name like '%St%'`
 - `select * from contacts where last_name like 'Po_tinger'` (`_` matches any one character)
 - `select * from contacts where lower(last_name) like 'st%'`
 - `select * from contacts where upper(last_name) like 'ST%'`
 - `select * from discounts where discount_message like '%%%'` (returns everything)
 - `select * from discounts where discount_message like '%%%' escape '!'` (returns everything that includes the string '%')
- You can compare against null with `is null` (= `NULL` does not work). You can negate with `not`.

3.7 Grouping and Ordering

- You can use functions like `upper` and dates when ordering.
- The `group by` keyword can be used to find unique data:

```
1 select status from orders group by status;
```

- By combining `group by` with `count` you can count the amount of unique data:

```
1 select status, count (*) from orders group by status;
```

- `group by` can also be used with the `where` keyword:

```
1 select name, count(*) as "Shipped Orders" from orders inner join
  customers using(customer_id) where status = 'Shipped' group by
  name order by "Shipped Orders" desc;
```

- `where` can NOT APPEAR AFTER `group by`; use the `having` keyword instead.
- The `having` keyword enables you to filter like with `where`, but after the `group by` keyword like so:

```
1 select status from orders where extract(year from order_date) > '
  2016' group by status having status like '%d';
```

- Multiple order by statements? First ordered by first statement, then “sub-ordered” by the second (last name the same -> now first name is evaluated).

- Want to have nulls first when ordering? Use nulls first or nulls last as the suffix.
- Removal of duplicates is done with `select distinct`. When multiple columns are being selected, use only one distinct keyword at the start. Multiple nulls are filtered (Null = Null).

3.8 Counting and Sums

- You can count the amount of rows with the `count()` function:

```
1 select count(*) from products
```

- The `sum` function can be used to calculate a total:

```
1 select sum(unit_price * quantity) from order_items;
```

- It can also be used to calculate a total per row (the group by `order_id` part is required; `group by order_value` does not work):

```
1 select order_id, sum(unit_price * quantity) as order_value from  
   order_items group by order_id;
```

3.9 Inserting

- It is a good idea to always specify the columns when inserting:

```
1 insert into discounts(discount_name, amount, start_date,  
   expired_date) values ('Summer Promotion', 9.5, date '2017-05-01',  
   date '2017-08-31')
```

- You can also “insert from select” using `insert into`:

```
1 insert into sales(customer_id, product_id, order_date, total)  
   select customer_id, product_id, order_date, sum(quantity *  
   unit_price) amount from orders inner join order_items using(  
   order_id) where status = 'Shipped' group by customer_id,  
   product_id, order_date;
```

- It's even possible to “create a table from select” using `create table x as`, basically coping its schema (`where 1 = 0` skips copying the rows):

```
1 create table sales_2017 as select * from sales where 1 = 0;
```

- Using `insert all`, it is possible to insert multiple rows at once (note the lack of commas between the into keywords. Here, the subquery is ignored/a placeholder.):

```
1 insert all into fruits (fruit_name, color) values ('Apple', 'Red')
  into fruits (fruit_name, color) values ('Orange', 'Orange')
  into fruits (fruit_name, color) values ('Banana', 'Yellow')
select 1 from dual
```

- You can also use conditions based on the subquery (`insert first` is the equivalent of a switch case.):

```
1 insert all when amount < 10000 then into small_orders when amount
  >= 10000 then into big_orders select order_id, customer_id, (
  quantity * unit_price) amount from orders inner join
  order_items using (order_id)
```

3.10 Switches

- Using `case` it is possible to create if/else constructs:

```
1 select product_name, list_price, case category_id when 1 then
  round(list_price * 0.05, 2) when 2 then round(list_price * 0.1,
  2) else round(list_price * 0.2, 2) end discount from products
```

- `case` is also useful for conditional grouping:

```
1 select * from locations order by country_id, case country_id when
  'US' then state else city end;
```

- `case` also evaluates to an expression, so you can use it for conditional updates:

```
1 update products set list_price = case when list_price < 20 then 30
  else 50 end where list_price < 50;
```

3.11 Helper Functions

- You can extract substrings with `substr`: `select substr('Alex', 1, 1) from dual;`
- Stuff like `select upper('uwu') from dual` can come in handy.
- Using `round` it is possible to round numbers (returns 5.23):

```
1 select round(5.234234234234, 2) from dual;
```

- You can use `replace` to replace strings:

```
1 update accounts set phone = replace(phone, '+1-', '');
```

- You can use the `floor`, `round` and `ceil` functions to get rounded values.

3.12 Auto-Generated Primary Keys

- `generated by default as identity` is quite useful for auto-incrementing columns such as PKs:

```
1 create table persons ( person_id number generated by default as
  identity, first_name varchar2(50) not null, last_name varchar2
  (50), primary key(person_id) );
```

- `generated always as identity` is the same but does not allow setting it manually.

3.13 Modifying Columns

- You can use `desc mytable` to show the schema for a table.
- `alter table` can be used to add columns using `add`:

```
1 alter table persons add birthdate date not null;
```

- You can also add multiples at once (note that there is no column keyword):

```
1 alter table persons add ( phone varchar2(20), email varchar2(100)
  )
```

- `modify` can change the column type (note that there is no column keyword):

```
1 alter table persons modify birthdate date null;
```

- `drop column` can be used to remove a column

```
1 alter table persons drop column birthdate;
```

- `rename column` can be used to rename a column:

```
1 alter table persons rename column first_name to forename;
```

- `rename to` can be used to rename a table:

```
1 alter table persons rename to people;
```

- `rename promotions to promotions_two` is an alternative syntax.

- You can use the `default` keyword to set a default value:

```
1 alter table accounts add status number(1,0) default 1 not null.
```

- A more efficient logical version of `drop column` is `set unused column`:

```
1 alter table suppliers set unused column fax;
```

- You can now drop it using:

```
1 alter table suppliers drop unused columns;
```

- If you want to physically drop a column, use `drop`:

```
1 alter table suppliers drop (email, phone);
```

3.14 Virtual Columns

- You can create virtual columns in regular tables without using views with `alter table x add ... as` (note the required `(` after the `as` keyword):

```
1 alter table parts add (capacity_description as ( case when
    capacity <= 8 then 'Small' when capacity > 8 then 'Large' end )
);
```

- The size of a `varchar2` is adjustable afterwards (note that this checks if any current `varchar2` s are larger than the new size and fails if they are.):

```
1 alter table persons modify first_name varchar2(255);
```

3.15 Modifying Tables

- You can drop a table with `drop table`:

```
1 drop table people;
```

- Appending `purge` clears the recycle bin; appending `cascade constraints` drop all related constraints.

- You can clear a table using `truncate table`:

```
1 truncate table customers_copy;
```

- The same limitations as with `drop table` concerning constraints apply, so appending `cascade (WITHOUT constraints)` drops all related ones.

- You can clear the recycle bin with:

```
1 purge recyclebin;
```

3.16 Constraints

- It is possible to add constraints (any constraints, a primary key in this example) after creating a table with `add constraint`:

```
1 alter table purchase_orders add constraint
  purchase_orders_order_id_pk primary key(order_id);
```

- You may remove a constraint with `drop constraint`:

```
1 alter table purchase_orders drop constraint
  purchase_orders_order_id_pk;
```

- Instead of removing it, you can also use `disable constraint`:

```
1 alter table purchase_orders disable constraint
  purchase_orders_order_id_pk;
```

- And re-enable it with `enable constraint`:

```
1 alter table purchase_orders enable constraint
  purchase_orders_order_id_pk;
```

- You can also add foreign key constraints:

```
1 alter table suppliers add constraint suppliers_supplier_groups_fk
  foreign key(group_id) references supplier_groups(group_id);
```

- Using a check constraint, arbitrary expressions can be evaluated:

```
1 alter table parts add constraint check_buy_price_positive check(
  buy_price > 0);
```

- A unique constraint prevents unwanted duplicates:

```
1 alter table clients add constraint unique_clients_phone unique(
  phone);
```

- With a not null constraint, fuzzy logic can be avoided; it is however best to define nullable fields at schema creation, as the syntax differs from the add constraint/drop constraint logic above:

```
1 alter table clients modify ( 7 phone not null );
```

- You can remove them by modifying it to null explicitly:

```
1 alter table clients modify ( phone null );
```

3.17 Types

- You can create a number within a range: `number(1,0)`.
- The `number` type is used for all types of numbers by specifying precision and scale: `number(6)` (or `number(6,0)`) is a signed integer fitting 6 digits, `number(6,2)` is a float with two digits precision. The DB doesn't just cut off numbers, it rounds them.
- The float type can be emulated by the number type, i.e. `float(2)` is equal to `number(38,2)`. The argument is in bits instead of digits though.
- The `lengthdb` function can be used to get the length of field in bytes.
- The char type has a fixed length: name `char(10)` or name `char(10 bytes)`, meaning that a char always takes up the amount of bytes set. `nchar` is the same but UTF-8 or UTF-16 any doesn't take bytes.
- The `varchar2` type also takes an argument for the length in bytes, which in ASCII corresponds to the amount of characters. `nvarchar2` is the same but UTF-8 or UTF-16 and doesn't take bytes.

3.18 Views

- You can create a view with `create view x as select ...`:

```
1 create view employees_years_of_service as select employee_id,
  first_name || ' ' || last_name as full_name, floor(
  months_between(current_date, hire_date) / 12) as
  years_of_service from employees;
```

- If used with `create or replace view`, upserts are possible.
- By appending `with read only`, you can prevent data modifications:

```
1 create or replace view employees_years_of_service as select
  employee_id, first_name || ' ' || last_name as full_name, floor(
  months_between(current_date, hire_date) / 12) as
  years_of_service from employees with read only;
```

- `drop view x` removes the view.
- Deletions and updates on views are usually fine, but inserts can often be not that useful due to fields being excluded from the view; see `instead of` triggers later on for a solution;
- Subqueries can be used in selects:

```
1 select * from ( select * from products) where list_price < 100;
```

- They can also be used in updates:

```
1 update ( select list_price from products ) set list_price =  
    list_price * 1.5;
```

3.19 Indexes

- You can create an index with `create index`:

```
1 create index members_last_name on members(last_name);
```

- You can also create an index spanning multiple columns:

```
1 create index members_full_name on members(first_name, last_name);
```

- You can drop an index with `drop index`:

```
1 drop index members_full_name;
```

4 PL/SQL

4.1 Block Structure

- Block structure:

```
1 declare  
2 -- declarations  
3 begin  
4 -- your logic  
5 exception  
6 -- exception handling  
7 end;
```

- The most simple example is as follows:

```
1 begin  
2     dbms_output.put_line('Hello World!');  
3 end;
```

- Use `put_line` from the `dbms_output` package to print to stdout.
- You can use the `declare` section for variables:

```
1 declare  
2     message varchar(255) := 'Hello, World!';  
3 begin  
4     dbms_output.put_line(message);
```

```
5 end;
```

- The `exception` block is used to handle exceptions, for example `zero_divide` for divisions by zero (when `others` then handles unexpected other exceptions):

```
1 declare
2     result number;
3 begin
4     result := 1/0;
5
6     exception
7         when zero_divide then
8             dbms_output.put_line(sqlerrm);
9         when others then
10            dbms_output.put_line('An unexpected error occurred: '
11                                || sqlerrm);
12 end;
```

- You always have to specify an execution section; use `null` for a no-op:

```
1 declare
2 begin
3     null;
4 end;
```

- You can use `--` for single line comments and `/*` for multi line comments.

4.2 Variables

- PL/SQL extends SQL by adding a boolean type (which can have the values true, false and null).
- Variables need not be given a value at declaration if they are nullable:

```
1 declare
2     total_sales number(15,2);
3     credit_limit number(10,0);
4     contact_name varchar2(255);
5 begin
6     null;
7 end;
```

- You can use `default` as an alternative to the `:=` operator when assigning variables in the declaration section. DO NOT use `=` when assignment, even re-assignment also uses `:=`.
- If a variable is defined as not null, it can't take a string of length 0:

```
1 declare
2     shipping_status varchar2(25) not null := 'shipped';
3 begin
```

```
4   shipping_status := ''; -- You need to specify any string != ''
5 end;
```

- Constants are created with the `constant` keyword and forbid reassignment:

```
1 declare
2     price constant number := 10;
3 begin
4     price := 20; -- Will throw an exception
5 end;
```

4.3 Fetching Data

- Use `select ... into` to fetch data into variables; `%TYPE` infers the type of a column:

```
1 declare
2     customer_name customers.name%TYPE;
3     customer_credit_limit customers.credit_limit%TYPE;
4 begin
5     select
6         name, credit_limit
7     into
8         customer_name, customer_credit_limit
9     from customers where customer_id = 38;
10
11     dbms_output.put_line(customer_name || ': ' ||
12                          customer_credit_limit);
12 end;
```

4.4 Branches and Expressions

- `if ... then ... end if` can be used for branching:

```
1 declare
2     sales number := 20000;
3 begin
4     if sales > 10000 then
5         dbms_output.put_line('Lots of sales!');
6     end if;
7 end;
```

- Inline expressions are also supported:

```
1 large_sales := sales > 10000
```

- Booleans need not be compared with `my_bool = true`, a simple `if my_bool then` is fine.

- `elseif ... then` is NOT valid syntax; `elsif ... then` is valid syntax.
- Statements may also be nested:

```
1 declare
2     sales number := 20000;
3 begin
4     if sales > 10000 then
5         if sales > 15000 then
6             dbms_output.put_line('A new sales record!');
7         else
8             dbms_output.put_line('Lots of sales!');
9         end if;
10    end if;
11 end;
```

4.5 Switches

- You may use the `case` keyword for switch cases:

```
1 declare
2     grade char(1);
3     message varchar2(255);
4 begin
5     grade := 'A';
6
7     case grade
8         when 'A' then
9             message := 'Excellent';
10        when 'B' then
11            message := 'Great';
12        when 'C' then
13            message := 'Good';
14        when 'D' then
15            message := 'Fair';
16        when 'F' then
17            message := 'Poor';
18        else
19            raise case_not_found;
20        end case;
21
22    dbms_output.put_line(message);
23 end;
```

4.6 Labels and Goto

- A `label/goto` equivalent is also available:

```
1  begin
2      goto do_work;
3      goto goodbye;
4
5      <<do_work>>
6      dbms_output.put_line('mawahaha');
7
8      <<goodbye>>
9      dbms_output.put_line('Goodbye!');
10 end;
```

4.7 Loops

- The equivalent of the **while** loop is the **loop**. **exit/continue** prevents an infinite loop:

```
1  declare
2      i number := 0;
3  begin
4      loop
5          i := i + 1;
6
7          dbms_output.put_line('Iterator: ' || i);
8
9          if i >= 10 then
10             exit;
11         end if;
12     end loop;
13
14     dbms_output.put_line('Done!');
15 end;
```

- For loops can be done using the **for i in 0..100 loop ... end loop** syntax:

```
1  begin
2      for i in 0..100 loop
3          dbms_output.put_line(i);
4      end loop;
5  end;
```

- While loops work as you'd expect; but also require the **loop** keyword:

```
1  declare
2      i number := 0;
3  begin
4      while i <= 100 loop
5          dbms_output.put_line(i);
6
7          i := i + 1;
```

```
8     end loop;
9 end;
```

4.8 Types and Objects

- You can also use `%ROWTYPE` to infer the type of a row and select an entire row at once:

```
1 declare
2     customer customers%ROWTYPE;
3 begin
4     select * into customer from customers where customer_id = 100;
5     dbms_output.put_line(customer.name || '/' || customer.website)
6         ;
7 end;
```

- It is also possible to use OOP-style object/row creation thanks to `%ROWTYPE`:

```
1 declare
2     person persons%ROWTYPE;
3
4 begin
5     person.person_id := 1;
6     person.first_name := 'John';
7     person.last_name := 'Doe';
8
9     insert into persons values person;
10 end;
```

4.9 Exceptions

- You can create custom exceptions:

```
1 declare
2     e_credit_too_high exception;
3     pragma exception_init(e_credit_too_high, -20001);
4 begin
5     if 10000 > 1000 then
6         raise e_credit_too_high;
7     end if;
8 end;
```

- If you want to raise a custom exception, use `raise_application_error`:

```
1 declare
2     e_credit_too_high exception;
3     pragma exception_init(e_credit_too_high, -20001);
```

```
4 begin
5     raise_application_error(-20001, 'Credit is to high!');
6 end;
```

- Using `sqlcode` and `sqlerrm` you can get the last exception's code/error message.

4.10 Cursors

- Using cursors, you can procedurally process data:

```
1 declare
2     cursor sales_cursor is select * from sales;
3     sales_record sales_cursor%ROWTYPE;
4 begin
5     update customers set credit_limit = 0;
6
7     open sales_cursor;
8
9     loop
10        fetch sales_cursor into sales_record;
11        exit when sales_cursor%NOTFOUND;
12
13        update
14            customers
15        set
16            credit_limit = extract(year from sysdate)
17        where
18            customer_id = sales_record.customer_id;
19    end loop;
20
21    close sales_cursor;
22 end;
```

- Complex exit logic can be avoided using the `for ... loop`:

```
1 declare
2     cursor product_cursor is select * from products;
3 begin
4     for product_record in product_cursor loop
5         dbms_output.put_line(product_record.product_name || ': $'
6                               || product_record.list_price);
7     end loop;
8 end;
```

- Cursors can also have parameters:

```
1 declare
2     product_record products%rowtype;
3     cursor
```

```
4      product_cursor (
5          low_price number := 0,
6          high_price number := 100
7      )
8  is
9      select * from products where list_price between low_price
10         and high_price;
11 begin
12     open product_cursor(50, 100);
13     loop
14         fetch product_cursor into product_record;
15         exit when product_cursor%notfound;
16
17         dbms_output.put_line(product_record.product_name || ' : $'
18                               || product_record.list_price);
19     end loop;
20     close product_cursor;
21 end;
```

4.11 Locks

- The DB can also lock fields for safe multiple access:

```
1 declare
2     cursor customers_cursor is select * from customers for update
3       of credit_limit;
4 begin
5     for customer_record in customers_cursor loop
6         update customers set credit_limit = 0 where customer_id =
7           customer_record.customer_id;
8     end loop;
9 end;
```

4.12 Procedures

- You can create procedures, which are comparable to functions:

```
1 create or replace procedure
2     print_contact(customer_id_arg number)
3  is
4     contact_record contacts%rowtype;
5 begin
6     select * into contact_record from contacts where customer_id =
7       customer_id_arg;
```

```
8      dbms_output.put_line(contact_record.first_name || ' ' ||  
                             contact_record.last_name);  
9  end;
```

- These procedures can then be executed:

```
1  begin  
2      print_contact(50);  
3  end;
```

- Or, without PL/SQL:

```
1  exec print_contact(50);
```

- Once a procedure is no longer needed, it can be removed with `drop procedure`:

```
1  drop procedure print_contact;
```

- It is also possible to infer a row type using `sys_refcursor` and return rows with `dbms_sql.return_result`:

```
1  create or replace procedure  
2      get_customer_by_credit(min_credit number)  
3  as  
4      customer_cursor sys_refcursor;  
5  begin  
6      open customer_cursor for select * from customers where  
                               credit_limit > min_credit;  
7  
8      dbms_sql.return_result(customer_cursor);  
9  end;
```

- You can now call it:

```
1  exec get_customer_by_credit(50);
```

4.13 Functions

- Functions are similar, but require returning a value:

```
1  create or replace function  
2      get_total_sales_for_year(year_arg integer)  
3  return number  
4  is  
5      total_sales number := 0;  
6  begin  
7      select sum(unit_price * quantity) into total_sales  
8      from order_items
```

```
9     inner join orders using (order_id)
10    where status = 'Shipped'
11    group by extract(year from order_date)
12    having extract(year from order_date) = year_arg;
13
14    return total_sales;
15 end;
```

- You can call them from PL/SQL:

```
1 declare
2     total_sales number := 0;
3 begin
4     total_sales := get_total_sales_for_year(2017);
5
6     dbms_output.put_line('Sales for 2017: ' || total_sales);
7 end;
```

- And remove them with `drop` function:

```
1 drop function get_total_sales_for_year;
```

4.14 Packages

- Packages can be used to group function “interfaces” and variables:

```
1 create or replace package order_management
2 as
3     shipped_status constant varchar(10) := 'Shipped';
4     pending_status constant varchar(10) := 'Pending';
5     cancelled_status constant varchar(10) := 'Canceled';
6
7     function get_total_transactions return number;
8 end order_management;
```

- You can now access the variables in the package with `..`:

```
1 begin
2     dbms_output.put_line(order_management.shipped_status);
3 end;
```

- In order to use functions in a package, you then have to create a package body, implementing it:

```
1 create or replace package body order_management
2 as
3     function get_total_transactions return number
4     is
```

```
5      total_transactions number;  
6      begin  
7          select sum(unit_price) into total_transactions from orders  
              inner join order_items using(order_id);  
8  
9          return total_transactions;  
10     end;  
11 end;
```

- You can now access the functions in the package with .:

```
1 select  
2     order_management.get_total_transactions() as  
        total_transactions  
3 from  
4     dual;
```

- And the same is possible from PL/SQL:

```
1 begin  
2     dbms_output.put_line(order_management.get_total_transactions()  
        );  
3 end;
```

- You can drop a package with **drop package** and a package body with **drop package body**:

```
1 drop package body order_management;  
2 drop package order_management;
```

4.15 Triggers

- Triggers follow a similar structure as procedures:

```
1 declare  
2 -- declarations  
3 begin  
4 -- your logic  
5 exception  
6 -- exception handling  
7 end;
```

- Using triggers, you can for example create a manual log after operations with **after update or delete on ...**:

```
1 create or replace trigger customers_audit_trigger  
2     after update or delete  
3     on customers
```

```
4   for each row
5 declare
6   transaction_type varchar2(10);
7 begin
8   transaction_type := case
9       when updating then 'update'
10      when deleting then 'delete'
11   end;
12
13   insert into audits(
14       table_name,
15       transaction_name,
16       by_user,
17       transaction_date
18   ) values (
19       'customers',
20       transaction_type,
21       user,
22       sysdate
23   );
24 end;
```

- Thanks to `before update of ... on ...`, it is also possible to do more complex checks before inserting:

```
1 create or replace trigger customers_credit_trigger
2   before update of credit_limit
3   on customers
4 declare
5   current_day number;
6 begin
7   current_day := extract(day from sysdate);
8
9   if current_day between 28 and 31 then
10      raise_application_error(-20100, 'Locked at the end of the
11         month');
12   end if;
13 end;
```

- In combination with `when`, `new` (not available in `delete` statements) and `old` (not available in `insert` statements), it is also possible to check based on the previous & current values:

```
1 create or replace trigger customers_credit_limit_trigger
2   before update of credit_limit
3   on customers
4   for each row
5   when (new.credit_limit > 0)
6 begin
7   if :new.credit_limit >= 2*:old.credit_limit then
8      raise_application_error(-20101, 'The new credit cannot be
```

```
9         more than double the old credit!');
10     end if;
11 end;
```

- Using `instead of` triggers and `returning ... into ...`, you can also use views to safely insert into multiple tables:

```
1 create or replace trigger create_customer_trigger
2   instead of insert on customers_and_contacts
3   for each row
4 declare
5   current_customer_id number;
6 begin
7   insert into customers(
8     name,
9     address,
10    website,
11    credit_limit
12  ) values (
13    :new.name,
14    :new.address,
15    :new.website,
16    :new.credit_limit
17  ) returning customer_id into current_customer_id;
18
19   insert into contacts(
20     first_name,
21     last_name,
22     email,
23     phone,
24     customer_id
25  ) values (
26    :new.first_name,
27    :new.last_name,
28    :new.email,
29    :new.phone,
30    current_customer_id
31  );
32 end;
```

- You can enable/disable a trigger with `alter trigger ... disable/enable`:

```
1 alter trigger create_customer_trigger disable;
```

- And completely remove it with `drop trigger`:

```
1 drop trigger create_customer_trigger;
```

- It is also possible to enable/disable all triggers of a table with `alter table ... enable/disable all triggers`:

```
1 alter table customers enable all triggers;
```

4.16 Maps

- Maps are also possible in PL/SQL using `table of`:

```
1 declare
2     type country_capitals_type
3         is table of varchar2(100)
4         index by varchar2(50);
5
6     country_capitals country_capitals_type;
7 begin
8     country_capitals('China') := 'Beijing';
9     country_capitals('EU') := 'Brussels';
10    country_capitals('USA') := 'Washington';
11 end;
```

- You can use `mymap.first` and `mymap.next` to iterate:

```
1 declare
2     type country_capitals_type
3         is table of varchar2(100)
4         index by varchar2(50);
5
6     country_capitals country_capitals_type;
7     current_country varchar2(50);
8 begin
9     country_capitals('China') := 'Beijing';
10    country_capitals('EU') := 'Brussels';
11    country_capitals('USA') := 'Washington';
12
13    current_country := country_capitals.first;
14
15    while current_country is not null loop
16        dbms_output.put_line(current_country || ': ' ||
17                               country_capitals(current_country));
18        current_country := country_capitals.next(current_country);
19    end loop;
20 end;
```

4.17 Arrays

- Using `varray`, it is also possible to create arrays:

```
1 declare
2     type names_type is varray(255) of varchar2(20) not null;
3
4     names names_type := names_type('Albert', 'Jonathan', 'Judy');
5 begin
6     dbms_output.put_line('Length before append: ' || names.count);
7
8     names.extend;
9
10    names(names.last) := 'Alice';
11
12    dbms_output.put_line('Length after append: ' || names.count);
13
14    names.trim;
15
16    dbms_output.put_line('Length after trim: ' || names.count);
17
18    names.trim(2);
19
20    dbms_output.put_line('Length after second trim: ' || names.
        count);
21
22    names.delete;
23
24    dbms_output.put_line('Length after delete: ' || names.count);
25 end;
```